

Context Free Grammar

What we have learnt so far

- DFA/NFA computers with extremely limited memory.
- Regular Expressions
Recipe to describe languages that are regular.

How about creating a slightly powerful computational model?

$$A = \{0^n 1^n \mid n \geq 0\}$$

Instead going directly to the new computational model, let's try to learn about something called context free grammar.

context free grammar to context free languages is what regular expressions are to regular languages.

☆ context free grammar is a collection of substitution rules also known as productions.

☆ Each rule appears as a line in the grammar.

☆ Each line comprises of a symbol, arrow, and a string.



☆ The string can consist of variables, and other symbols called terminals.

☆ Variables are represented using capital letters, terminals are represented using lowercase letters, numbers or special symbols.

☆ One variable is designated as the starting variable.

we know that following language A is not regular.

$$A = \{ 0^n 1^n \mid n \geq 0 \}$$

$$\Sigma = \{ 0, 1 \}$$

1. $S \longrightarrow 0S1$

2. $S \longrightarrow B$

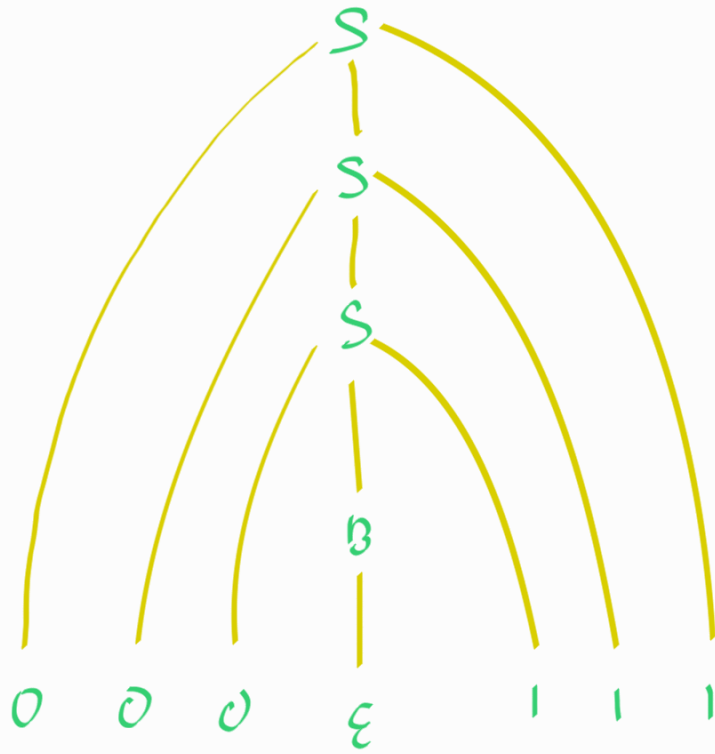
3. $B \longrightarrow \epsilon$

$$\begin{array}{ccccccc} S & \xRightarrow{(1)} & 0S1 & \xRightarrow{(1)} & 00S11 & \xRightarrow{(1)} & 000S111 \\ & & & & & & \downarrow (2) \\ & & & & & & 000111 \\ & & & & & \xleftarrow{(3)} & 000B111 \end{array}$$

The sequence of substitutions to obtain a string is called a derivation

$$S \xRightarrow{(2)} B \xRightarrow{(3)} \epsilon$$

we can represent this using a parse tree as well.



$$S \rightarrow 0S1$$

$$S \rightarrow B$$

$$B \rightarrow \epsilon$$

000ε111

Any language that can be generated by some context free grammar is called a context free language.

Just like regular expressions and regular languages

The language created by a context free grammar G is denoted as $L(G)$

Definition

A context free grammar is a 4-tuple (V, Σ, R, S) , where

1. V is the set of **variable**.
2. Σ is a finite set, called terminals.
3. R is a set of rules, with each rule being a variable and a string of variables and terminals.
4. S is the starting variable.

Example

$$G = (\{S\}, \{a, b\}, R, S)$$

$$R: 1. S \longrightarrow a S b$$

$$2. S \longrightarrow S S$$

$$3. S \longrightarrow \varepsilon$$

$$S \longrightarrow a S b \mid S S \mid \varepsilon$$

$$\begin{array}{ccccccc} S & \xRightarrow{1} & a S b & \xRightarrow{2} & a \underline{S} S b & \xRightarrow{3} & a \varepsilon S b & \xRightarrow{2} & a S S b \\ & & & & & & & \Downarrow 1 & \\ & & & & & & & & a a b b S b \\ & & & & & & & \xleftarrow{2} & a a \underline{S} b S b \\ & & & & & & & \xleftarrow{2} & a a b S b \\ & & & & & & & \xleftarrow{2} & a a b b \end{array}$$

$$G_2 = (V, \Sigma, R, \langle \text{Expr} \rangle)$$

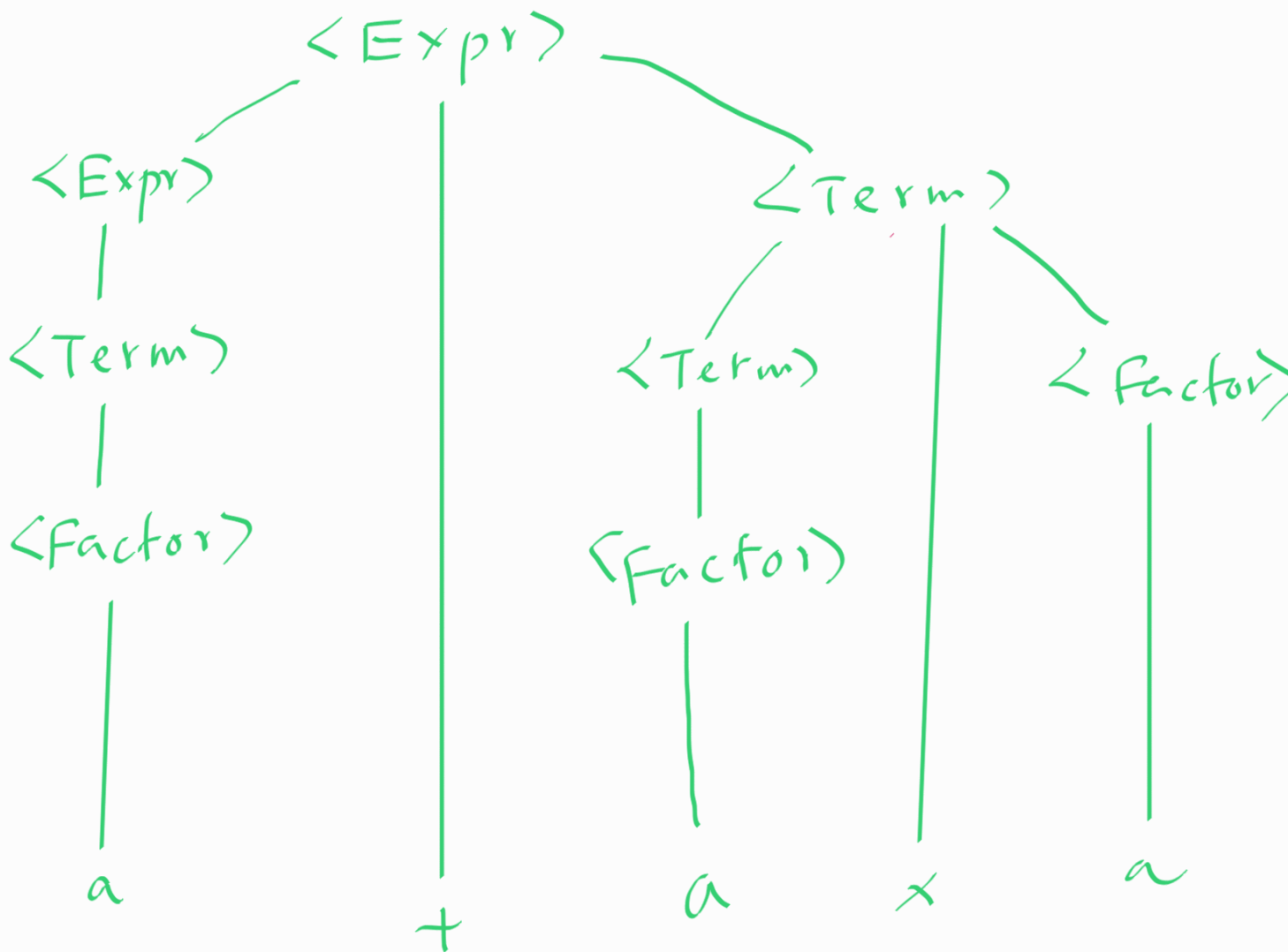
$$V = \{ \langle \text{Expr} \rangle, \langle \text{Term} \rangle, \langle \text{Factor} \rangle \}$$

$$\Sigma = \{ a, +, \times, (,) \}$$

$$1. \langle \text{Expr} \rangle \rightarrow \langle \text{Expr} \rangle + \langle \text{Term} \rangle \mid \langle \text{Term} \rangle$$

$$2. \langle \text{Term} \rangle \rightarrow \langle \text{Term} \rangle \times \langle \text{Factor} \rangle \mid \langle \text{Factor} \rangle$$

$$3. \langle \text{Factor} \rangle \rightarrow (\langle \text{Expr} \rangle) \mid a$$



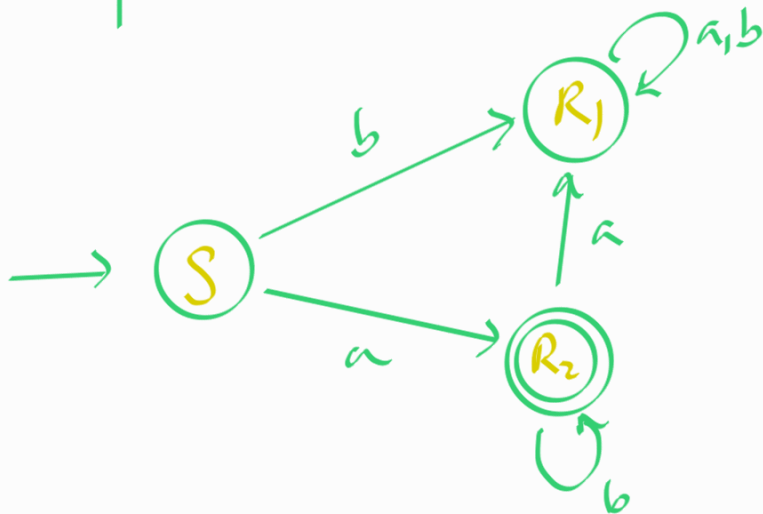
$$a + a \times a$$

Side Note!

we can easily represent any regular language using context free grammar.

Suppose you have a DFA, you can easily convert it to a CFG.

Example: DFA $\longrightarrow$ Context Free Grammar



* make a variable for each state q_i

* Create a grammar rule for each transition in the DFA

* Use the variable that you created for the start state as your start variable

* Add $R_i \rightarrow \epsilon$ for each accept state R_i

$$S \longrightarrow b R_1$$

$$S \longrightarrow a R_2$$

$$R_1 \longrightarrow a R_1 \mid b R_1$$

$$R_2 \longrightarrow b R_2 \mid a R_1$$

$$R_2 \longrightarrow \epsilon$$